8. Data Cleaning and Preparation

Current raw-file status

The supplied CSV has 100,000 rows and 37 columns. There are 0 duplicate full rows and 0 duplicate vehicle IDs. There are 9,511 missing cells overall. This document reports the raw file profile; it does not claim that the CSV has already been transformed.

Preparation plan

1. Preserve `ev_market_test.csv` unchanged.
2. Import using a parser that preserves numeric values and empty cells as missing.
3. Confirm field units and source definitions.
4. Validate `vehicle_id` uniqueness.
5. Review category spelling and whitespace.
6. Investigate extreme numeric values before deciding whether they are errors.
7. For Tableau, set numeric and categorical data roles correctly.
8. Choose metric-specific null handling; do not drop all rows because one unrelated field is null.
9. Record every exclusion, recoding, or calculation.
10. Compare pre/post row counts and key summaries.

Null handling recommendations

- `battery_cycle_life`: leave null or exclude only from analyses requiring cycle life unless source supports a defensible imputation.
- `resale_value_pct_3yr`: exclude nulls from resale analysis and report denominator.
- `annual_sales_units`: exclude nulls from sales analysis; inspect extreme values before using averages.
- `customer_rating`: exclude nulls from rating summaries; do not assume a missing rating equals zero.

Transformations to avoid without justification

- Capping outliers just to improve chart appearance.
- Replacing missing values with zero.
- Treating record count as unit sales.
- Combining regions or battery types without documenting the mapping.